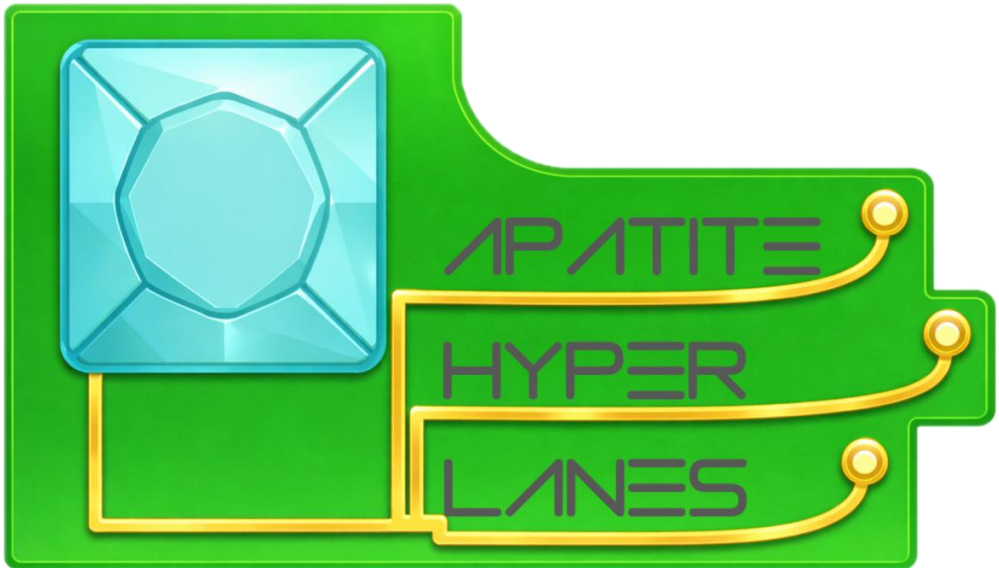


AHL's PC proposal for BTB's Request (03-2026a)

Date:
April 15, 2026

Prepared for:

BTB Heavy Equipment, Inc

PO Box 1520
Saskatoon, SK S7K 3R5
(306) 659 – 4256
(306) 659 – 4267

Prepared by:

Apatite Hyper Lanes, Ltd

Regina, SK S4P 3A3
(306) 123 – 12345
apatitehyperlanesbiz@ahl.ca

Any inquiries with respect to the proposal may be directed to:

Peter Gane, Team Lead / Communications Manager | gane7896@saskpolytech.ca

Kolten Degelman, Tech Lead | degelman9745@saskpolytech.ca

Ruby Hennig, Lead Analyst | hennig3326@saskpolytech.ca

Summary

Technical Summary

The computer that Apatite Hyper Lanes (AHL) proposes within this document can run all relevant work software for the majority of BTB Heavy Equipment's employees. Additionally, the computer is easy to upgrade if the need arises. The design is lightweight and small, good for uncertainty in an office environment. This document contains detailed descriptions of each part used to build the computer, and arguments for why AHL chose those parts. There is also a budget section that details the prices of all parts used in the proposed computer, and the prices for buying all 250 copies of each part.

Executive Summary

Described within this document are detailed explanations for why this computer is the right choice for BTB Heavy Equipment. The formal introduction of the project will begin to describe the benefits of choosing AHL as a computer provider. In the project description there will be details for each component and why they are a good choice. The CPU is both affordable and can keep up with modern CPUs in terms of functionality. The motherboard and case combination is small and can fit wherever it is needed in the office, with a built-in power supply and cooling system. The mouse and keyboard are well reviewed and have a sleek design that is both stylish and modern. Overall, the computer will fulfill all duties required of it in this office setting. An outline of the costs and some benefits will be found in the budget section. Further details can be found about the decision making and benefits of proposal within the conclusion. Figures, definitions, and details about the claims and contents of the document will be in the appendices.

Table of Contents

AHL's PC proposal for BTB's Request (03-2026a) 0
Summary i
Table of Contents ii
Table of figures iv
Formal introduction 1
Project description 2
Introduction 2
CPU 3
Motherboard 5
RAM 8
Storage 10
Optical Drive 11
Card Reader 12
Case + PSU 13
Monitor 15
Keyboard 16
Mouse 17
Headset 18
Conclusion 20
Conclusion 21
Budget 22
References 23
Appendices 26
Appendix A: Glossary 26
Appendix B: Communications 27
In person communications 29
Appendix C: Amendments 30

Appendix D: Specifications sheet 31
Appendix E: Supporting Data 33

Table of figures

Budget 1 22
Communications Figure 1 27
Communications Figure 2 27
Communications Figure 3 28
Communications Figure 4 28
Communications Figure 5 29
Specifications sheet 1 31
Supporting Evidence figure 1 33
Supporting Evidence figure 2 33
Supporting Evidence figure 3 34

Formal introduction

Apatite Hyper Lanes has reviewed the specifications and worked with the BTB Heavy Equipment systems administrators to put together a proposal that will meet the client's needs and stay within budget. What follows is the recommended solution.

The computers that have been selected provide a strong base for the BTB office. The machines can run Microsoft applications such as Word, Excel, and PowerPoint, as well as accounting, browser, and networking software. These machines are easy to implement and use, so anyone can start working quickly and comfortably. The parts have been chosen because they are versatile and easy to upgrade. If some team members require more powerful capabilities, such as rendering 3D models, these computers can be expanded with additional RAM and graphics cards as needed.

Shipment and production will begin as soon as Apatite Hyper Lanes has been selected for this project, the parts will arrive at the Apatite warehouse where inventory can be taken. When read a trained team of builders will be able to complete production within two weeks and deliver your computers within a healthy timeframe.

Apatite Hyper Lanes is confident these machines will meet your needs while staying within budget.

Project description

Introduction

In the description below there will be an outline for a machine that will be a good base in an office setting. The parts below have an emphasis on compactness and functionality and will work well for BTB. Cuts have been made in favour of a computer that will thrive no matter the size of the office as you will be able to tell by our choice of both case and motherboard. The system contains the fundamentals of memory, storage and processing power. The keyboard, mouse, and headset have been chosen for their comfortable and easy to use nature that will support work in the office.

CPU



The CPU being used is the Intel Core i3-8100 3.6 GHz Quad-Core (see Table 1, CPU section, p.32). The requirements in the BTB request document (Johnson & Khawaja p.2) specified a "Dual core" with "At least 2.5 GHz" and "Must match motherboard slot/socket type." (Johnson & Khawaja p.2). During internal discussions, market analysis indicated that certain constraints relied on discontinued platforms, resulting in limited supply and higher costs. A request was submitted to Khawaja, Yahya, to revise the CPU requirements (Communications figure 1, p.27), and after review, the request was approved (Communications figure 2, p.27). This amendment relaxed the hard requirement for a fixed dual-core processor to a soft two core minimum.

The i3-8100 manufactured by Intel is based on their Coffee Lake core architecture that utilizes four physical cores and four processing threads (see Table 1, CPU section, p.32). The i3-8100 features a 6 MB cache and operates using 65W of power (see Table 1, CPU section, p.32). Supporting up to 64 GB of dual channel DDR4-2133MHz memory, the processor connects to the motherboard using the LGA1151 socket (see Table 1, CPU section, p.32). The CPU sourced from Amazon (see Table 1, CPU section, p.32) includes integrated Intel UHD Graphics 630, and a stock cooling solution (see Table 1, CPU section, p.32).

During our initial discussions it was determined that the dual-core requirement listed was problematic and presented limitations. After market analyses, the decision came to would be to contact BTB's tech lead to discuss possible amendments that could be made, resulting in the removal of the hard limit, lifting the dual-core requirement. The expanded range of viable CPU options allowed the selection of a more suitable component.

Based on this information, the i3-8100, with its four cores operating at a clock speed of 3.6 GHz, exceeds both the two core minimum requirement and the 2.5 GHz minimum requirement. The i3-8100 using the LGA1151 socket is compatible with the chosen motherboard and its dual-channel capability (see Table 1, Motherboard section, p.32). The CPU's RAM support covers the chosen RAM type and speed (see Table 1, Memory section, p.32). This CPU configuration provides a balanced level of performance and efficiency for the intended use case. The chosen CPU also satisfies several additional listed requirements; it includes a stock CPU cooler fan (Communications figure 1, p.27) which reduces additional system cost and complexity, and the requirement for a "Video card – or integrated" (Communications figure 1, p.27) is covered by the integrated Intel UHD Graphics 630. These supplemental features provide a very cost-effective means to cover several non-CPU requirements.

With the selected intel processor as the CPU its four-core processor has a major performance boost over the previous generations dual-core architecture. "Adding two more physical cores... about a third faster" (Hexus), this makes the CPU far more capable in multitasking and modern software environments. It has strong single-core performance and its market value is very competitive. Being of the new quad-core platform it is comparable to the previous generations mid-ranged CPUs while still being of low cost, "on par or beating a Core i5 Skylake processor" (TechSpot, Intel Core i3-8100 specs, n.d.) "a decent low-end entry" (Phoronix). with good value. The choice to go with Intel as the manufacturer was because its single-core performance is superior than its competitors like AMD* "almost always faster than the Ryzen 3 1300X" (TechSpot, Intel Core i3-8100 specs, n.d.). Its stock setting performs competitively against the alternatives as "the Core i3-8100 beats the Ryzen 3 1200 in AVX workloads" (Tom's Hardware).

Motherboard



The motherboard being used is the ASRock B360-ITX/ac Mini ITX LGA1151 (see Table 1, Motherboard section, p.32). The requirements specified “Must have correct slot - memory - speed - dual-channel – minimum 4 USB – IEEE 1394 port – WOL if NIC is not integrated – Windows 11 compatible” (Johnson & Khawaja p.2). The requirements in related sections also specified the need for Aux port, a VGA port for the monitor, and ECC capabilities to run ECC capable RAM (Johnson & Khawaja p.2). During internal discussion market analysis found that the original constraints included outdated features, no longer supported by current products to be found, which resulted in limited supply, and higher costs. Some of the requirements were also found to be unclear. A request was submitted to Khawaja, Yahya to revise the motherboard requirements (Communications figure 1, p.27), which was approved (Communications figure 2, p.27). An in-person discussion with Abitha Fenson was had, who clarified the requirements (Appendices C I, IV, V, p.32). The amendments changed the requirement to have need an IEEE 1394 “fire wire” port, eliminated the requirement for VGA making any video port compatible with the monitor acceptable, the ECC requirements were cut, and the necessary number of Aux ports was clarified as needing just one out port for the speakers.

The B360-ITX Mini is manufactured by ASRock, sourced from Newegg (Needs citation) and uses the mini-ITX form factor for compact cases, includes a PCIe expansion slot, and an NVMe M.2 slot for additional storage along with four SATA ports for internal drives (see Table 1, Motherboard section, p.32). The board comes with intel 802.11ac WiFi, and the NIC uses Intel Gigabit LAN firmware supporting the RJ-45 port (see Table 1, Motherboard section, p.32). The LGA 1151 socket supports the chosen CPU, and the DDR4-2133 DIMM slots support the selected RAM, which has dual-channel architecture for running two RAM modules in dual-channel mode (see Table 1, Motherboard section, p.32).

To support external devices and peripherals, the back panel includes multiple USB ports, a PS/2 port for legacy input devices, and three audio jacks for analog sound in & output (see Table 1, Motherboard section, p.32). For display support, it provides DisplayPort, HDMI, and DVI connectors, allowing compatibility with different monitor types (see Table 1, Motherboard section, p.32).

While discussing which motherboard would fulfil all the criteria, it was determined that the need for many of the included requirements did not fit well with the PC's use-case and many of the requirements came across quite vague. Since the monitor was to be supplied by AHL the strict need to include VGA was seen as unnecessary. The need for ECC capabilities was also questioned as ECC RAM is generally server grade and expensive. The cost of a ECC capable motherboard alone was costing more than the entire budget, so it was decided to cut the ECC capability. When attempting to fulfill the requirement for the legacy Fire Wire port, it was deemed impossible to procure an in-production motherboard that came with Fire Wire natively. We discussed adding an expansion card but decided that including the Fire Wire port, was beyond the scope of the PC's use case (Communications figure 3, p.28). There was initial uncertainty regarding the necessary amount of Aux ports as the requirement didn't specify if the procured headset needed to be analog or not, but after a discussion with a system administrator it was clarified that the headset port did not have a set requirement.

Based on this information the ASRock B360-ITX/ac fulfils these requirements by being equipped with the LGA1151 socket, matching the selected CPU, it supports DDR4 memory compatible with the chosen RAM, including its speed and has dual-channel capabilities satisfying the architecture specification. The motherboard has the following pertinent ports, four USB, an HDMI, a LAN RJ-45, and an out Aux jack, fulfilling the USB, monitor, LAN, and Aux, requirements. The motherboard also satisfies several other requirements as it comes with an integrated NIC, covering the NIC requirements. The motherboard provides enough USB slots for the included peripherals; the mini ITX form factor ensures that the board fits within the case dimensions, and it has four SATA ports

which sufficiently covers the included internal devices (see Table 1, Optical Drive, Card Reader, & SSD sections, p. 32 & 33).

We selected the ASRock motherboard over other boards as the ASRock is extra durable, increasing the life of components with its 60A Power Choke and Dual-Stack MOSFET (ASRock). The integrated WiFi add flexibility meaning the PC can still be used in rooms without cable access (ASRock). Despite being a mid-range chipset, the board "includes Wi-Fi capabilities, USB 3.1 (10 Gbps)... as well as a M.2 slot" (MBReview). As a professional board, it is "a good value for any buyer who wants a small non-overclocking motherboard." (MBReview).

RAM



The RAM being used is the Corsair CMV8GX4M1A2133C15 8 GB (1 x 8 GB) DDR4-2133 memory module (see Table 1, Memory section, p.32). The RAM's specification is "Minimum 8 GB, single DIMM – ECC capabilities", and it must be compatible with the selected motherboard and CPU (Johnson & Khawaja p.2). During internal discussion and market analysis it was determined that the specified ECC capabilities needed to be reviewed. A request was submitted to Khawaja, Yahya to revise the ECC requirements (Communications figure 1, p.27), which was approved (Communications figure 2, p.27). This resulted in the chosen RAM not requiring ECC capabilities.

The 8 GB RAM stick is manufactured and shipped by Corsair and uses the DDR4 generation of RAM (see Specifications sheet, Memory section, p. 35) which matches the motherboard's DIMMs (see Specifications sheet, Motherboard section, p. 35). The total capacity of the RAM is 8 GB through one stick (see Table 1, Memory section, p.35). The Corsair RAM has a speed of 2133MHz, and a latency of CL15, while the operating voltage is 1.2V (see Table 1, Memory section, p.32).

During the preliminary discussions on which RAM would best meet the use case, the required ECC had limited the selection to a small selection of either DDR3 RAM, which was in short supply, or expensive server grade RAM, so it was decided to cut the ECC requirement.

Based on this information it was decided that the Corsair RAM fulfilled the requirements because it uses a single module meeting the 8 GB requirement with the single DIMM preference. The speed and timings of the RAM are supported by the motherboard and CPU (see Table 1, CPU and Motherboard sections, p32) satisfying the compatibility requirement.

The Corsair RAM was chosen due to its competitive value in the office environment "designed for standard computing situations where performance is not the primary concern" (Corsair). Its performance is not compromised by its low cost and is "carefully screened and tested at the factory" (Corsair). It was also a perfect fit for this build that uses an Intel CPU as it has "guaranteed compatibility with Intel platforms" (Corsair). "DDR4 reduces power consumption by operating at 1.2V" (Crucial) making it an ideal choice over DDR3 platforms. The more economical speed bracket which the RAM uses is also ideal as "faster memory has limited impact on general computing performance" (TechSpot, DDR4 memory performance scaling, 2016). One 8 GB stick over two 4 GB sticks is well suited for future flexibility as only one additional 8 GB module is required to double the systems RAM instead of replacing two sticks, the motherboards "dual-channel memory can significantly improve bandwidth" (TechSpot, DDR4 memory performance scaling, 2016).

Storage



The storage solution chosen is the Silicon Power A55 512 GB 2.5" Solid State Drive (see Table 1, Storage section, p.33). The requirements for the storage found in the BTB documents are for a “Hard drive… 500 GB, attaches to… SATA cable”. (Johnson & Khawaja p.2). In the preliminary meeting with Abatha Fenson the requirement for a hard drive was replaced with a solid-state drive (SSD).

The 512 GB 2.5” SSD is manufactured by Silicon Power and mounts the 3.5” bay within a 2.5” adapter (see Table 1, Storage section, p.33) into the chosen case (see Table 1, Case + PSU section, p.33). The SSD connects to the motherboard with a SATA data cable transferring data at 530+ MB/s and is powered by the power supply unit (PSU) using a SATA power cable (see Table 1, Storage section, p.33). This provides the PC with 512 GB of storage space for data (see Table 1, Storage section, p.33). The SSD gets shipped from Silicon Power (see Table 1, Storage section, p.33).

For cost and simplicity, the decision was made to use a SATA SSD over an NVMe SSD. Based on this information the Silicon Power SSD satisfies the requirement for 500 GB of space having 512 and still uses a SATA cable like in the original specification. As an SSD the Silicon power 2.5” drive fulfills the need for a solid-state drive, while fitting into the internal 3.5” drive bay in the chosen case and is powered using a SATA power cable from the power supply (see Table 1, Case + PSU section, p.33).

The decision to use the Silicon Power A55 SSD was due to its cost compared to an NVMe “SATA SSDs offer a balance between cost and performance” (TechSpot, SSD vs HDD vs NVMe comparison, 2018), while still providing excellent speeds “SSDs are significantly faster than HDDs in boot times and file access” (Crucial) operating close to the theoretical limit of SATA III (~600 MB/s) (Silicon). Its 2.5” slim form factor ensures compatibility with mini cases.

Optical Drive



The optical drive chosen is the Internal SATA DVD±RW Optical Drive (see Table 1, Optical Drive section, p.33) to meet the requirements to “support installation of software and the creation of backups” (Johnson & Khawaja p.2) specified in the BTB document.

The Optical Drive is a generic DVD drive sold and shipped by Nixsys, with read write capabilities, it mounts to an external 5.25” drive bay and accepts standard CDs and DVDs (see Table 1, Optical Drive section, p.33). The drive connects to the motherboard with a SATA data cable and is powered by the PSU using a SATA power cable (see Table 1, Optical Drive section, p.33).

Using this information, it was decided that the internal SATA drive provided by Nixsys satisfied the criteria for an optical drive that can install software and create backups with its read write capabilities. It mounts to the 5.25” external bay on the chosen case (see Table 1, Case + PSU section, p.33), connecting to the motherboard with a SATA data cable while getting power from the PSU using a SATA power cable (see Table 1, Case + PSU section, p.33).

Choosing to go with the Nixsys provider was because it provided the lowest cost solution on the market as the majority of sellers were asking for substantially more for a generic optical drive (see Supporting Evidence fig 1.1). We went with an internal drive over an external USB drive as they are typically cheaper and reduces setup complexity (TechTarget).

Card Reader



The Card reader chosen is the EnLabs Metal Case 3.5" Front Panel Card Reader (see Table 1, Card Reader section, p.33) for satisfying the requirements “Must support multiple types including CompactFlash™.” (Johnson & Khawaja p.2) specified in the BTB document.

The 3.5” card reader is manufactured by ENLabs and sourced from Newegg, mounting into an external 3.5” drive bay while connecting to the motherboard with a 9-pin USB 2.0 header, providing both power and data (see Table 1, Card Reader section, p.33). The media support that the front panel card reader adds is for MS, CF/MD, SD/MMC, XD, T-F, and USB 2.0, all transferring data at 480 Mbps (see Table 1, Card Reader section, p.33).

The EnLabs 3.5” external card reader supports all card devices stated in the requirements doc including CompactFlash which is among its seventeen different supported data transfer types. Mounting to an external 3.5” bay, it fits the external 3.5” bay found in the chosen case (see Table 1, Case + PSU section, p.33). Using a USB 2.0 9-pin header for both power and data makes it compatible with the selected motherboard (see Table 1, Motherboard section, p.32).

The EnLabs card reader is a cost-effective solution to the need for a card reader as it supports more than just CompactFlash, offering “seventeen-in-one… easily read and write to your digital memory cards” (Newegg listing). Unlike external card readers, it provides easy front access and eliminates cable clutter, reducing setup complexity (retail consultant).

Case + PSU



The case chosen is an Apex MI-008 Mini ITX Tower Case w/250 W power Supply (see Table 1, Case + PSU section, p.33) which also contains a built-in PSU. The requirements of the case are as follows “Mini-tower, able to hold all other components.” (Johnson & Khawaja p.2) with the requirements of the power supply being “450 W minimum. Must match motherboard connector type.” (Johnson & Khawaja p.2). During internal discussions it was determined that to best meet the size and power draw of the PC while better aligning with the current market environment, necessary changes should be made to the power supply requirements. A communication was sent to the system administrators with suggestions to move forward with (Figure 2.1, p. 29). The response confirmed the changes resulting in the minimum wattage of the power supply to dynamically meet the needs of the PC rather than being static (Communications figure 2, p.27).

The MI-008 Case and power supply made by Apex and sold by Amazon (see Table 1, Case + PSU section, p.33) is made to support mini-ITX motherboards in the tower case profile; the dimensions of the case are 5.10”x 8.70”x 13.50” holding an internal volume equaling 8.5 liters (see Table 1, Case + PSU section, p.33). The mini tower case front panel includes two USB ports and two Aux ports (see Table 1, Case + PSU section, p.33), and comes with three different drive bays, a 5.25” external, one being a 3.5” external, and an internal 3.5” drive bay (see Table 1, Case + PSU section, p.33). The included rear mounted power supply is of a generic original equipment manufacturer (OEM) build and supplies 250 watts of power (see Table 1, Case + PSU section, p.33). The PSU is a SFX small form factor make and is ATX12V compliant, meaning it comes with a 24-pin ATX motherboard connector, a 4-pin CPU connector, two SATA power connectors, three 4-pin Molex connectors and lastly a floppy power connector (see Table 1, Case + PSU section, p.33).

Early drafts of the PC contained the same case (see Supporting Evidence fig 2.1) with an additional PSU to be swapped with the OEM PSU. This added great cost, and the benefit of the additional wattage was deemed minimal to nonexistent. It was decided to send an email inquiry regarding the issue arguing that 250W was still well over the industry standard 30% surplus (Seasonic). The response gave the go ahead to use the new 30% metric when deciding which PSU to choose.

The Apex case is of the mini-ITX standard accepting the form factor of the selected motherboard (see Table 1, Motherboard section, p.32), while satisfying the requirement for a mini ITX tower. The three drive bays found in the case suitably accommodate the selected optical drive (see Table 1, Optical Drive section, p.33), the selected card reader (see Table 1, Card Reader section, p.33), and the selected SSD (see Table 1, Storage section, p.33). This makes the case ideal for retaining all the components of this pc, satisfying its need in the requirements. The OEM 250W PSU provides over 30% surplus power to the build which at typical operation draws and estimated 110W of power (see Supporting Evidence fig 3.1). This makes the OEM option ideal for the required power consumption while also driving down cost, and waste.

The choice to go with the Apex MI-008 was multifold, its size most of all makes it “a great choice for instances where you don’t have a lot of space for a system” (Newegg listing). Its “Sturdy construction…” that’s “…Thermally optimized…” with “…Excellent EMI shielding…” holding “…8.5 liters” (Newegg) in its compact volume makes it the ideal choice when selecting for the mini tower form factor. Additionally, it’s “equipped with a 250W power supply unit to handle typical power consumption for most Mini-ITX systems” (Newegg) so the plan to go without the auxiliary PSU has virtually no impact on the PC’s function. The case also has a practical flexible orientation able to be used in a horizontal desktop-style or vertical tower-style (Newegg). Despite is small size it incredibly has both a 5.25” and 3.5” drive with space inside for a 3.5” internal bay and an expansion slot for future potential customization.

Monitor



The chosen monitor is the Acer EK241Y Gbi 24-inch IPS 1080p 120 Hz monitor (see Table 1, Monitor section, p.33). The BTB document specifies the monitor must be a “17-inch or better monitor, supports 1024×768 at 75Hz or better, best if it supports video card’s maximum resolution” (Johnson & Khawaja p.2).

The 24-inch monitor manufactured by Acer includes an HDMI cable, and is sourced from Amazon (see Table 1, Monitor section, p.33) featuring a 1920 x 1080 (Full HD) display running at a 120Hz refresh rate (see Table 1, Monitor section, p.33). The monitor’s stand uses an ergonomic tilt design and can also be removed to be attach to a wall mount (see Table 1, Monitor section, p.33).

The tech team contacted the lead analyst for further research into the market. As a result, the EK241Y was found to out preform all previously considered options (Shoprbc). With this information it was decided that the Acer EK241Y monitor fulfils the requirements with its 24” screen exceeding the 17” minimum, the 120Hz refresh rate also surpasses the 75Hz minimum set. It comes with its own HDMI cable to plug into the available HDMI port on the motherboard (see Table 1, Motherboard section, p.32) which simplifies procurement. The integrated graphics on the selected CPU can also display at the monitors maximum native resolution (see Table 1, CPU section, p.32) satisfying the requirement that the video output supports the monitor’s maximum resolution.

The 24” Acer monitor “presents stunning, high-quality images with excellent detail…”, “…120Hz refresh rate… delivers smooth 2D motion scenes…”, and “…displays color accurately up to 178 degrees” (Acer, n.d.). The stand comes with ergonomic “tilt from -5° to 20°” (Acer, n.d.) but can be detached for a wall mount providing versatility. It has a great resolution and refresh rate for its price point, out preforming monitors of lesser quality (Shoprbc).

Keyboard



The Adesso AKB-160UB Wired keyboard (see Table 1, Keyboard section, p.33) was chosen to satisfy the keyboard requirements in the BTB document which are “Ergonomic, with standard PS/2 or USB connector.” (Johnson & Khawaja p.2) was needed.

The AKB-160UB keyboard sourced from PC-Canada uses a split ergonomic design with an integrated palm rest meant to keep the wrists at a healthy angle (see Table 1, Keyboard section, p.33). It has low profile quiet membrane keycaps and uses a wired USB connector (see Table 1, Keyboard section, p.33).

During internal discussions, there was confusion relating to the definition of ‘ergonomic’. A meeting was held with a systems administrator where the question was brought up, and it was clarified as to the exact meaning of ‘ergonomic’.

Using this reasoning with the Adesso ergonomic wired keyboard’s split design, the ergonomic requirement is satisfied. The USB connector can be plugged into one of the open USB ports on the back of the motherboard (see Table 1, Motherboard section, p.32) covering the need for a specific wired connection.

The Adesso AKB-160UB full-sized keyboard with its “quiet membrane keys… lifecycle of more than five million keystrokes” (Adesso, n.d.)helps maintain lower ambient office noise levels. Its “wired USB connection ensures reliable performance” (Staples) and doesn’t need batteries to operate. The ergonomic design “encourages natural hand positioning… for maximum comfort” (Adesso, n.d.) which “reduces typing fatigue during long use” (Staples).

Mouse



The chosen mouse is the Logitech B100 Wired Optical Mouse (see Table 1, Mouse section, p.33), the requirements of the mouse listed in the BTB document are “Either PS/2 or USB connector, wheel optical mouse preferred.” (Johnson & Khawaja p.2).

The wired mouse is manufactured by Logitech and sourced from Vuugo (see Table 1, Mouse section, p.33). It’s connected using a 1.8 m cable length USB and features both optical tracking and a scroll wheel (see Table 1, Mouse section, p.33). The mouse utilizes three buttons and is ambidextrous (see Table 1, Mouse section, p.33).

Using this information, it was decided that the Logitech B100 with its scroll wheel, satisfies the requirement to include one. Its USB connector has an open port on the back of the motherboard (see Table 1, Motherboard section, p.32) with a cable length of 1.8 m, fulfilling the connector qualification. The mouse uses optical tracking, covering the requirement. It is ambidextrous insuring that everyone can use it.

The B100 is an ideal office and general use mouse as its“plug-and-play… just plug it in… get back to business” (Logitech). The “comfortable… ambidextrous design” (Logitech) is ideal for a diverse staff force, and the “wired USB connection ensures reliable performance” (Best Buy) without needing batteries. Its low cost doesn’t negatively impact its quality as its “built… with high quality and reliability standards” (Logitech).

Headset



The Mad Catz F.R.E.Q 4 7.1 Channel (see Table 1, Headset section, p.33) is the headset chosen to satisfy the BTB requirements “Jacks for microphone and speakers. Headset must have noise cancellation” (Johnson & Khawaja p.2). During internal discussions there was some uncertainty around the connector required for the headset and the definition of ‘noise cancelling’, an in-person discussion was held with a system administrator (In person communications, paragraph 3, p.31) resolving these issues.

The wired headset, sourced from Amazon (see Table 1, Headset section, p.33) is manufactured by Mad Catz and uses a USB connector to plug into the pc; it comes with a retractable microphone using environmental noise cancellation (ENC) and the suspension headband with large ear-cups provide a noise cancelling seal (see Table 1, Headset section, p.33). The headset features 50mm neodymium drivers and 7.1 surround-sound (see Table 1, Headset section, p.33).

The requirement for the speakers, microphone and headset to all use Jacks was found to be an unclear necessity, as well as the exact definition of ‘noise cancellation’ was found to be quite subjective which led to uncertainty. An in-person meeting was held, and it was clarified that the headset could use a USB connection instead of Aux exclusively. It was also clarified that ‘noise cancelling’ referred to any headset advertising it (In person communications, paragraph 3, p.31).

With the following informing the decision came to was that the Mad Catz F.R.E.Q 4 with its USB connector can connect to the available USB port on the back of the motherboard (see Table 1, Motherboard section, p.32) which avoids interfering with the speakers’ need for Aux ports. The retractable microphone using ENC and the large sealing

ear-cups supported by a suspension headband provides ample noise cancellation. The 50mm drivers also results in high quality clear sound.

The F.R.E.Q 4 has “super-sized 50mm Neodymium drivers… deliver remarkable bass and crisp highs” (Mad Catz). Proving “clear and loud audio” (TechRadar). With its “suspension headband…” and “oversized ear-cups…” it “provide exceptional comfort” (Mad Catz) “comfortable to wear… for long periods of time” (TechRadar). Mad Catz ENC microphone “picks up every word… while blocking external audio” (Mad Catz), great for voice calls and with it being retractable it can be stored when not in use.

Conclusion

As has been demonstrated each part was selected with consideration for both their utility and price. Apatite Hyper Lanes has presented these parts with utmost care and has ensured each part will be compatible with each other. Some of the parts were changed slightly from the original requirements document and these changes can be found detailed below in the appendix section. If further upgrades are necessary for some types of work these can be accomplished by changing the RAM, adding storage in the NVMe slot, or by using the PCIe slot to add a video card or something of such.

Conclusion

To conclude this proposal, this computer is optimized for BTB's needs when accounting for the requirements and the limited budget. Any adjustments made to the requirements were necessary in reaching this goal and are available to view in the Appendices section. AHL is a skilled company of passionate computer builders who want to provide an excellent service to their customers. Each computer is made with attention to detail suited to each customer's needs.

In the case of the proposal, it was decided that a small lightweight computer would help in a new office building where everyone is trying to get settled. Employees unfamiliar with technology will find that the computer is approachable and easy to understand. With a compact but powerful design this is the most versatile and cost-effective solution that can be found within the boundaries of the request.

Although this is a good computer all around there are many changes that can be made after the computers have been purchased. Apatite Hyper Lanes has created plans for several changes that can be made to each computer to specialize them for the unique circumstances of the workers and the tasks they are assigned.

Budget

The price for each part at time of creation of this document is listed below in the centre column and the right column will list the total cost when all 250 parts are purchased.

Budget 1

	Price per part	Total Part price
CPU	$143.03	$35,757.50
Motherboard	$183.08	$45,770.00
RAM	$91.99	$22997.50
Storage	$101.50	$25,375.00
Optical Drive	$55.00	$13,750.00
Card Reader	$19.29	$4,822.50
Case	$175.50	$43,875.00
Monitor	$89.99	$22,497.50
Mouse	$12.44	$3,110.00
Keyboard	$56.99	$14,247.50
Headset	$34.99	$8,747.50
Total	$967.65	$241,912.50

The computers will be equipped with all the office standard specifications required for operations. A small and sturdy build it will be able to fit where it is best suited for each employee's personal workspace. Future compatibility has been ensured with each of the modern components. The monitors have an excellent framerate that will be comfortable on the eyes of the employees. The ergonomic mouse and keyboard are well recommended and tested for relaxed working conditions. The team of AHL is confident in a design that can be changed if the need arises to compensate for sudden market changes.

References

Acer. (n.d.). *EK241Y Gbi monitor specifications*. Retrieved from https://www.acer.com/us-en/monitors/essential/ek1/pdp/UM.QE1AA.G05

Adesso. (n.d.). *TruForm ergonomic desktop keyboard (AKB-160UB)*. Retrieved from https://adesso.com/shop/solutions/corporate-solutions/akb-160ub/

Amazon. (n.d.). *Acer EK241Y Gbi 24in IPS FHD 1920 x 1080 120Hz 1ms Frameless Gaming*. Retrieved from https://www.amazon.ca/EK241Y-Gbi-Frameless-Freesync-Ergonomic/dp/B0F79PCTVG?adgrpid=1349101000567473&hvadid=84319324334504&hvbmt=be&hvdev=c&hvlocphy=124061&hvnetw=o&hvqmt=e&hvtargid=kwd-84318982776328%3Aloc-32&hydadcr=1502_13523021&mcid=29305100184230b

Amazon. (n.d.). *ASRock B360M-ITX/AC LGA1151/Intel B360/DDR4/SATA3&USB3.1/M.2/Wi-Fi/GbE/Mini-ITX Motherboard*. Retrieved from https://www.amazon.ca/ASRock-B360M-ITX-AC-Mini-ITX-Motherboard/dp/B07BMVHFBW

Amazon. (n.d.). *Intel Core i3-8100 Desktop Processor*. Retrieved from https://www.amazon.ca/dp/B0759FTRZL?language=en_CA&th=1

Amazon. (n.d.). *Mad Catz The Authentic F.R.E.Q. 4 Gaming Headset – Black : Amazon: Video Games*. Retrieved from https://www.amazon.ca/dp/B084X7JNBC?language=en_CA&th=1

ASRock. (n.d.). *B360M-ITX/ac motherboard specifications*. Retrieved from https://www.asrock.com/mb/Intel/B360M-ITXac/index.asp

Canada, Best Buy. (n.d.). *Logitech B100 wired mouse listing*. Retrieved from https://www.bestbuy.ca/en-ca/product/logitech-b100-optical-usb-corded-mouse-910-001439/12397636

Corsair. (n.d.). *ValueSelect 8GB DDR4 2133MHz memory module*. Retrieved from https://www.corsair.com/us/en/p/memory/cmv8gx4m1a2133c15/value-select-8gb-1-x-8gb-ddr4-dram-2133mhz-c15-memory-module-cmv8gx4m1a2133c15

Crucial. (n.d.). *DDR4 memory overview*. Retrieved from https://www.crucial.com/articles/about-memory/ddr4-overview

Crucial. (n.d.). *SSD vs HDD comparison*. Retrieved from https://www.crucial.com/articles/about-ssd/ssd-vs-hdd

Hexus. (2017). *Intel Core i3-8350K & i3-8100 Coffee Lake review*. Retrieved from https://m.hexus.net/tech/reviews/cpu/115481-intel-core-i3-8350k-core-i3-8100-14nm-coffee-lake/

Logitech. (n.d.). *B100 optical USB mouse specifications*. Retrieved from https://www.logitech.com/en-ca/products/mice/b100-usb-business-mouse.html

Mad Catz. (n.d.). *F.R.E.Q 4 gaming headset specifications*. Retrieved from https://www.madcatz.com/En/Product/Detail/freq4

MBReview. (n.d.). *ASRock B360M-ITX/ac overview*. Retrieved from https://www.mbreview.com/news/motherboard-review-database/6655/asrock-b360m-itx-ac/

Newegg. (n.d.). *Apex MI-008 Mini-ITX case product listing*. Retrieved from https://www.newegg.ca/apextechusa-mini-itx-tower-mi-series-steel-computer-case-black-mi-008/p/N82E16811154091

Newegg. (n.d.). *EnLabs U2CR35MA card reader product listing*. Retrieved from https://www.newegg.ca/en-labs-u2cr35ma/p/0DS-00JP-00006

Phoronix. (2017). *Intel Core i3-8100 Linux performance review*. Retrieved from https://www.phoronix.com/review/core-i3-8100

Retail Consultant. (n.d.). *EnLabs U2CR35MA product page*. Retrieved from https://retailconsultant.ca/p/enlabs-u2cr35ma-metal-case-3-5-floppy-bay-front-panel-sdhc-micro-sd-mmc-cf-xd-tf-flash-memory-internal-all-in-1-usb-2-0-card-reader-w-usb-port-hub/

Seasonic. (n.d.). *How to use the PSU calculator? guide in 2025 for PC geeks. How to Use the PSU Calculator? Guide in 2025*. Retrieved from https://seasonic.com/insights/psu-calculator-using-guide-2025/

shoprbc. (n.d.). *Acer v226hql HBI 22-inch FHD Monitor. shopRBC.com :: Acer V226HQL Hbi 22-inch FHD Monitor*. Retrieved from https://www.shoprbc.com/ca/shop/product_details.php?pid=58910465

Silicon Power. (n.d.). *A55 solid state drive specifications*. Retrieved from https://www.silicon-power.com/product-detail/ace_a55/

Staples. (n.d.). *Adesso ergonomic USB keyboard product listing*. Retrieved from https://www.staples.ca/products/3135549-en-adesso-ergonomic-usb-keyboard-wired-canadian-bilingual-layout-black

TechRadar. (2020). *Mad Catz F.R.E.Q 4 review*. Retrieved from https://www.techradar.com/reviews/mad-catz-freq-4

TechSpot. (2016). *DDR4 memory performance scaling*. Retrieved from https://www.techspot.com/article/1171-ddr4-ram-speed-performance/

TechSpot. (2018). *SSD vs HDD vs NVMe comparison*. Retrieved from https://www.techspot.com/guides/1635-ssd-vs-hdd-vs-nvme/

TechSpot. (n.d.). *Intel Core i3-8100 specs*. Retrieved from https://www.techspot.com/specs/cpu/174550-intel-core-i3-8100.html

TechTarget. (n.d.). *Optical storage definition*. Retrieved from https://www.techtarget.com/whatis/definition/optical-storage

Tom's Hardware. (2017). *Intel Core i3-8100 CPU review*. Retrieved from https://www.tomshardware.com/reviews/intel-core-i3-8100-cpu-review,5385-8.html

Verna, J., & Yahya, K. (2026). *Request for Proposal (03-2026a)*.

Appendices

Appendix A: Glossary

ECC: Error Correcting Code is a feature supported by some compatible hardware to prevent data corruption.

System Administrator: The system administrators of BTB Heavy Equipment are Yahya Khawaja, and Abitha Fenson.

Intel: A company that manufactures microprocessors and chipsets.

Core Architecture: The CPU's design.

Amazon: A worldwide shipping company.

Socket: connector on the motherboard to the CPU.

AMD: A company that manufactures microprocessors and chipsets.

Aux: Analog audio auxiliary port.

VGA: Video Graphics Array is a video port.

ASRock: A company that manufactures hardware such as motherboards.

Case: The physical shell to keep the components of the computer in.

PCIe expansion slot: Port on the motherboard for adding extra functionality.

NVMe slot: Port on the motherboard for adding additional solid state storage units.

Silicon Power: A company that manufactures computer storage drives.

Nixsys: A company that supplies components.

EN-Labs: A company that manufactures card readers.

Apex: A company that manufactures computer cases.

Acer: A company that manufactures various computer components.

PC-Canada: A company that supplies computer parts.

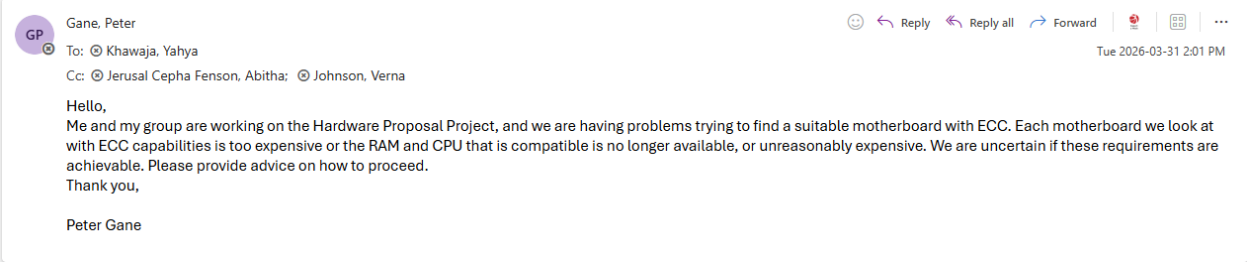
Logitech: A company that manufactures computer parts.

Mad Catz: a company that manufactures audio output devices.

Appendix B: Communications

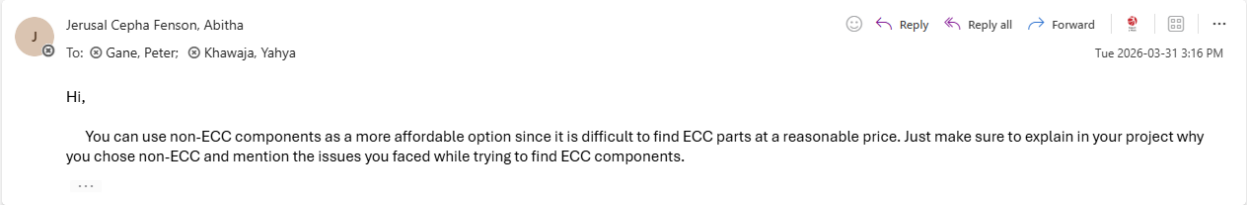
Communications Figure 1

Email to Systems administrator asking for exemption of ECC.



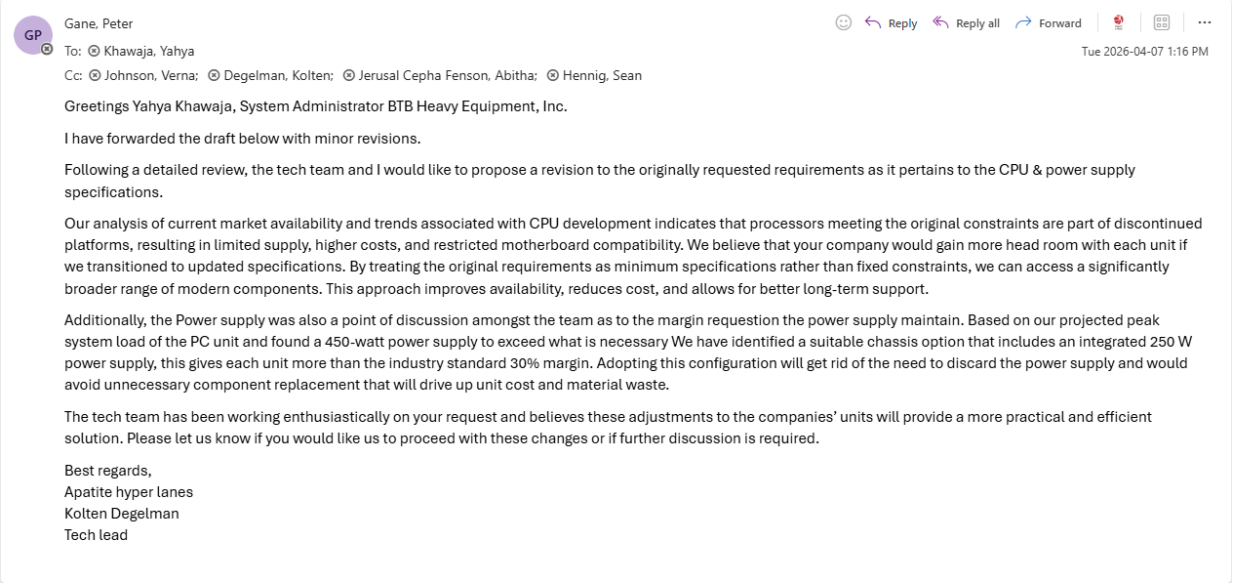
Communications Figure 2

Email from systems administrator confirming we do not need ECC.



Communications Figure 3

Email To systems administrator requesting a different CPU requirement and less wattage for PSU.



GP

Gane, Peter

Reply | Reply all | Forward

To: Khawaja, Yahya

Tue 2026-04-07 1:16 PM

Cc: Johnson, Verna; Degelman, Kolten; Jerusal Cepha Fenson, Abitha; Hennig, Sean

Greetings Yahya Khawaja, System Administrator BTB Heavy Equipment, Inc.

I have forwarded the draft below with minor revisions.

Following a detailed review, the tech team and I would like to propose a revision to the originally requested requirements as it pertains to the CPU & power supply specifications.

Our analysis of current market availability and trends associated with CPU development indicates that processors meeting the original constraints are part of discontinued platforms, resulting in limited supply, higher costs, and restricted motherboard compatibility. We believe that your company would gain more head room with each unit if we transitioned to updated specifications. By treating the original requirements as minimum specifications rather than fixed constraints, we can access a significantly broader range of modern components. This approach improves availability, reduces cost, and allows for better long-term support.

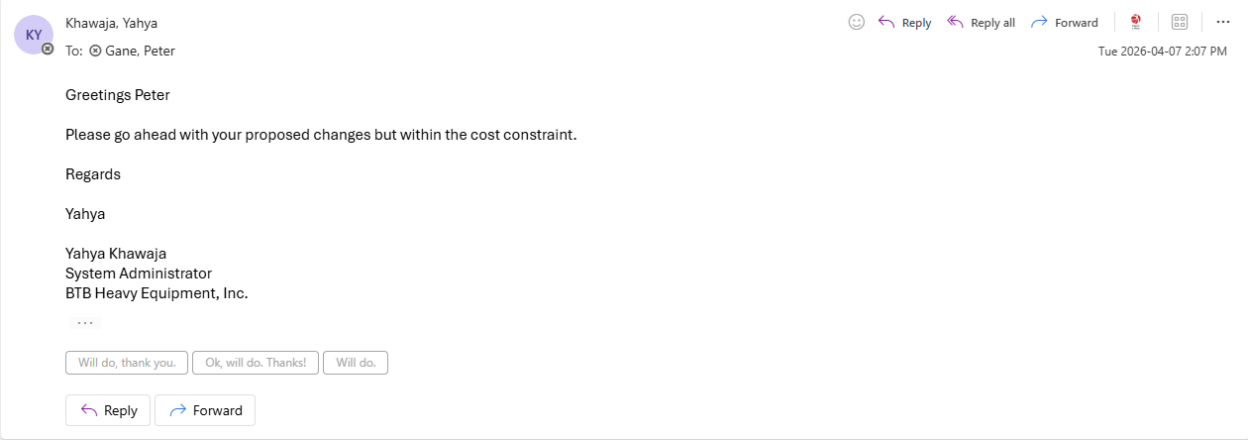
Additionally, the Power supply was also a point of discussion amongst the team as to the margin requestion the power supply maintain. Based on our projected peak system load of the PC unit and found a 450-watt power supply to exceed what is necessary We have identified a suitable chassis option that includes an integrated 250 W power supply, this gives each unit more than the industry standard 30% margin. Adopting this configuration will get rid of the need to discard the power supply and would avoid unnecessary component replacement that will drive up unit cost and material waste.

The tech team has been working enthusiastically on your request and believes these adjustments to the companies' units will provide a more practical and efficient solution. Please let us know if you would like us to proceed with these changes or if further discussion is required.

Best regards,
Apatite hyper lanes
Kolten Degelman
Tech lead

Communications Figure 4

Email from systems administrator confirming the component changes.



KY

Khawaja, Yahya

Reply | Reply all | Forward

To: Gane, Peter

Tue 2026-04-07 2:07 PM

Greetings Peter

Please go ahead with your proposed changes but within the cost constraint.

Regards

Yahya

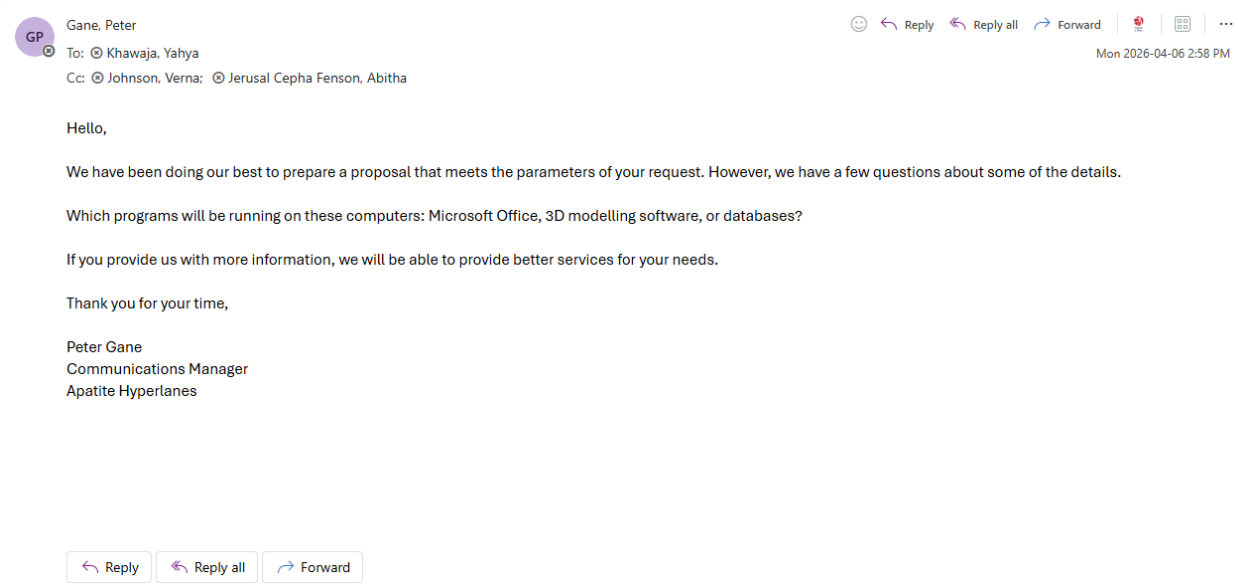
Yahya Khawaja
System Administrator
BTB Heavy Equipment, Inc.

Will do, thank you. | Ok, will do. Thanks! | Will do.

Reply | Forward

Communications Figure 5

Email to systems administrator asking the purpose of the computers



In person communications

In addition to the email communications, we had a meeting with the systems administrator to discuss whether the VGA port and firewire requirements were necessary. The conclusion was that they could be excluded from the build.

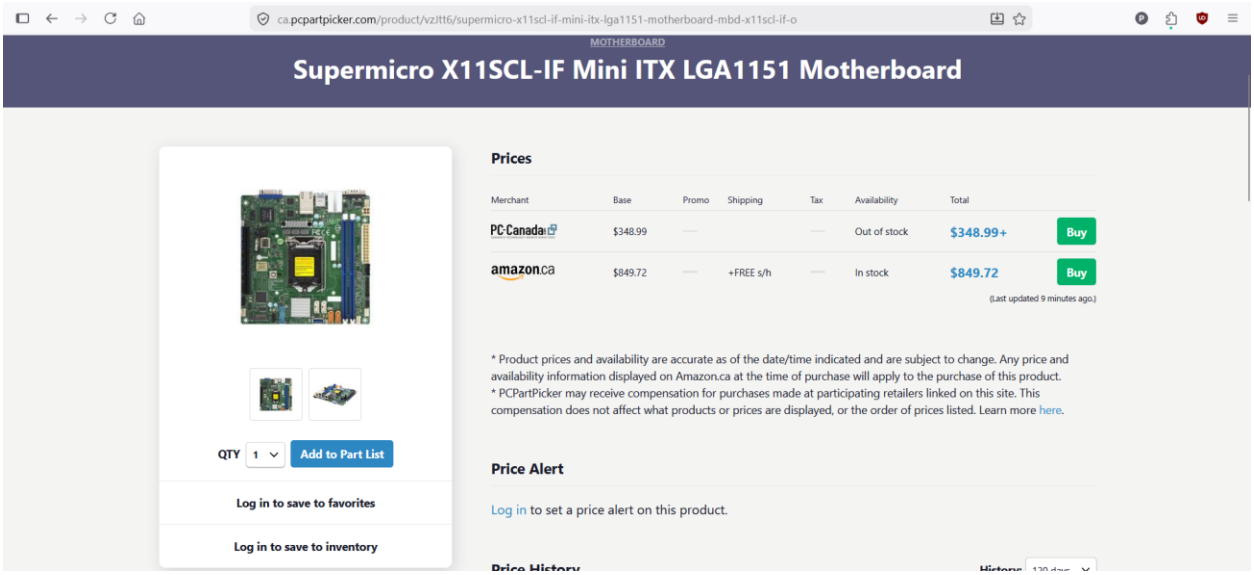
There was another meeting regarding if SSD or Hard drive was the preferred storage type. The conclusion was that SSD would be used.

Finally, a meeting was held to discuss if headphones needed to be connected by USB or aux and it was decided either was fine. In the same meeting we discussed the client's definition of noise canceling.

Appendix C: Amendments

As mentioned, there are components that do not follow the original requests. Detailed below are the reasons for why each exemption was requested.

I. ECC – ECC is not included in this proposal as we deemed it to be too expensive and difficult to acquire for the motherboard form factor needed for the case. If ECC was considered necessary or very beneficial it may have been requested that a different type of case was used however there will be no major issues if ECC is not included. Find in the picture below the other found motherboard in the same form factor as the final one.



II. CPU – The CPU requirement was read as though it had to be a two-core processor. Two-core processors are currently obsolete and are no longer manufactured in large quantities. As such a confirmation was sent to ensure that the CPU could be upgraded beyond two-core.

III. Power Supply Unit – The Power supply unit in the requirements needed to have a capacity of 400W. It is recommended that the power capacity for a PSU is 30% more than the total required capacity of all the parts required wattage put together. Since 400W far exceeds the value of all the parts plus 30%, a weaker PSU was chosen

IV. Firewire – Firewire is no longer supported by most operating systems. It was deemed to be too costly in both the budget and in how it might affect the rest of the part's compatibility and thus excluded.

V. VGA – The VGA port is an outdated video connection standard and is used less frequently in cheap and easier to obtain video cards.

Appendix D: Specifications sheet

Specifications sheet 1

Table 1 Spec Sheet			
Part	*Name*	*Cost per*	*Source*
CPU	Intel Core i3-8110 3.6 GHz Quad-Core Processor	$134.27	https://tinyurl.com/wh8bn6xx
Specs	4 Cores | 4 Threads | 3.6GHz PBF | 6MB Cache | LGA1151 Socket | 64GB Max Memory | DDR4-2400 Memory Types | Dual-Channel | Stock Cooler | Coffee Lake Core Architecture | 65W TDP | Intel UHD Graphics 630 Integrated Graphics | 4096x2304 Max Resolution		
Motherboard	ASRock B360-ITX/ac Mini ITX LGA1151 Motherboard	$183.08	https://tinyurl.com/pk84zrrm
Specs	Mini-ITX Form Factor | LGA1151 Socket | 32GB Max memory | 2x DDR4 DIMM | Dual-Channel | DDR4-2666/2400/2133 Memory Types | 1xPCIe x16 | 1x NVMe M.2 | 4x SATA6.0 | 1x PCU Fan(4-pin) | 1x Chassis Fan(4-pin) | 1x USB2.0 Header | 1x USB3.1 Gen1 Header | Intel UHD Graphics Support | 4096x2160 Max Resolution | Intel 802.11ac WiFi | Intel Gigabit LAN NIC | 1x RJ-45, | 2x USB3.1 Gen2 | 2x USB3.1 Gen1 | 1x PS/2 | 1x DVI-I | 1x HDMI | 1x DP1.2 | 3x Aux		
Memory	Corsair CMV8GX4M1A2133C15 8 GB DDR4-2133 CL15 Memory	$91.99	https://tinyurl.com/23cvc6wb
Specs	8GB(1x8GB) Memory Size | 15-15-15-36(CL 15) Latency |		

	DDR4 DIMM Memory Type | 2133MHz Speed | 1.2 Voltage		
Storage	Silicon Power A55 512 GB 2.5" Solid State Drive	$101.50	https://tinyurl.com/ybvedk87
Specs	512GB Capacity | 2.5” Form Factor | SATA6.0 Connector		
Optical drive	Internal SATA DVD±RW Optical Drive (5.25-inch)	$55.00	https://nixsys.com/dvd-rw-drive
Specs	DVD±RW | CD-RW | 5.25" Form Factor | SATA6.0 Connector		
Card reader	EnLabs U2CR35MA Metal Case 3.5"Floppy Bay Front Panel	$19.29	https://tinyurl.com/bdzpb562
Specs	Compact Flash | Memory Stick | MMC | SD | SmartMedia | xD | USB2.0 Connector | 3.5” Form Factor | 480Mbps Speed		
Case + PSU	Apex MI-008 Mini ITX Tower Case w/250 W power Supply	$175.50	https://tinyurl.com/58d5ek8v
Specs	Mini-ITX Tower Form Factor case | Mini-ITX Form Factor MB | 2x USB2.0 Front Panel | 2x Aux Front Panel | 8.5L Capacity | 8.7"x5.1"x13.5" Dimensions | 1x 5.25" External | 1x 3.5" External | 1x 3.5 Internal | ATX12V PSU Standard | SFX Form Factor | 250W Supply		
Monitor	Acer EK241Y Gbi 24in IPS FHD 1920 x 1080 120Hz	$89.99	https://tinyurl.com/y54ecc7n
Specs	1920x1080 Resolution | HD, 120Hz Refresh Rate | 16:9 Aspect Ratio | LED Display Type | 23.8" Screen Size| 6lb Weight | HDMI Port | VGA Port | Ergo Tilt | Wall Mount Capable | 1x HDMI cable | 1x Power cord		
Keyboard	Adesso AKB-160UB Wired Ergonomic Keyboard	$56.99	https://tinyurl.com/6hua59jr
Specs	Full-sized | qwerty Layout | Membrane Switches | 1.83m Wired USB Connector | Split Layout | Palm Rest | English(US) | 5 Million Keystroke Life		
Mouse	Logitech B100 Wired Optical Mouse	$12.44	https://tinyurl.com/bde6f749
Specs	Ambidextrous | 1.8m Wired USB Connector | Optical | 800dpi | 3x buttons | Scroll Wheel		
Headset	Mad Catz F.R.E.Q 4 7.1 Channel Headset	$34.99	https://tinyurl.com/yunrkshm
Specs	Over Ear | 7.1 Surround-Sound | Sound Isolation Noise Cancelation | 50mm Neodymium Driver | Retractable ENC microphone | suspension headband | Wired USB Connector | 0.84g Weight		

Acer. (n.d.). EK241Y Gbi monitor specifications. Retrieved from https://www.acer.com/us-en/monitors/essential/ek1/pdp/UM.QE1AA.G05

Adesso. (n.d.). TruForm ergonomic desktop keyboard (AKB-160UB). Retrieved from https://adesso.com/shop/solutions/corporate-solutions/akb-160ub/

ASRock. (n.d.). B360M-ITX/ac motherboard specifications. Retrieved from https://www.asrock.com/mb/Intel/B360M-ITXac/index.asp

Logitech. (n.d.). B100 optical USB mouse specifications. Retrieved from

https://www.logitech.com/en-ca/products/mice/b100-usb-business-mouse.html

Mad Catz. (n.d.). F.R.E.Q 4 gaming headset specifications. Retrieved from https://www.madcatz.com/En/Product/Detail/freq4

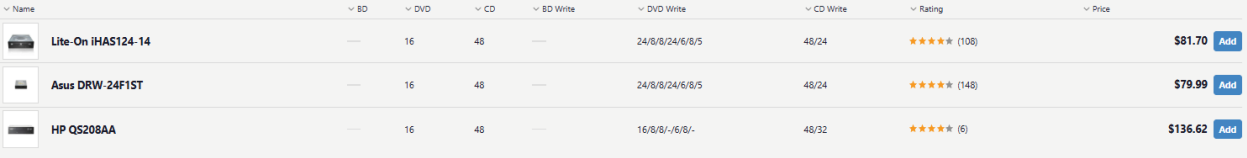
Silicon Power. (n.d.). A55 solid state drive specifications. Retrieved from https://www.silicon-power.com/product-detail/ace_a55/

TechSpot. (n.d.). *Intel Core i3-8100 specs*. Retrieved from https://www.techspot.com/specs/cpu/174550-intel-core-i3-8100.html

Appendix E: Supporting Data

Supporting Evidence figure 1

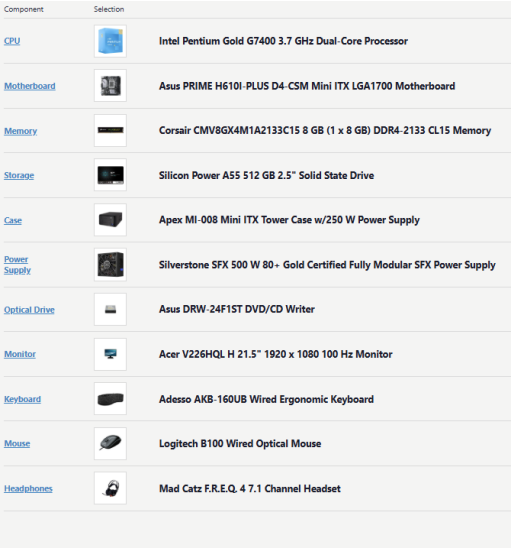
Expensive average price of Optical Drives



Name	BD	DVD	CD	BD Write	DVD Write	CD Write	Rating	Price	
Lite-On iHAS124-14	—	16	48	—	24/8/8/24/6/8/5	48/24	★★★★★ (108)	$81.70	Add
Asus DRW-24F1ST	—	16	48	—	24/8/8/24/6/8/5	48/24	★★★★★ (148)	$79.99	Add
HP QS208AA	—	16	48	—	16/8/8/-/6/8/-	48/32	★★★★★ (6)	$136.62	Add

Supporting Evidence figure 2

First full build draft



Component	Selection	
CPU		Intel Pentium Gold G7400 3.7 GHz Dual-Core Processor
Motherboard		Asus PRIME H610I-PLUS D4-CSM Mini ITX LGA1700 Motherboard
Memory		Corsair CMV8GX4M1A2133C15 8 GB (1 x 8 GB) DDR4-2133 CL15 Memory
Storage		Silicon Power A55 512 GB 2.5" Solid State Drive
Case		Apex MI-008 Mini ITX Tower Case w/250 W Power Supply
Power Supply		Silverstone SFX 500 W 80+ Gold Certified Fully Modular SFX Power Supply
Optical Drive		Asus DRW-24F1ST DVD/CD Writer
Monitor		Acer V226HQL H 21.5" 1920 x 1080 100 Hz Monitor
Keyboard		Adesso AKB-160UB Wired Ergonomic Keyboard
Mouse		Logitech B100 Wired Optical Mouse
Headphones		Mad Catz F.R.E.Q. 4 7.1 Channel Headset

Supporting Evidence figure 3

(PC power consumption calculated by PC Part Picker.com

https://ca.pcpartpicker.com/list/TWdLJw) 110 + (110 * 0.3) = 143 < 250

